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Department of **Electronics and Communication Engineering**

Course: Digital Communication and Networks
Course Code:21EC63

Project Report on

Securing a Remote Branch Office - Configuring a secure connection between a main office and a remote branch office, protecting data transfer from unauthorized access..

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Introduction

In an increasingly interconnected world, businesses often have multiple office locations spread across different geographical regions. These remote branch offices need to securely communicate and exchange data with the main office. However, this interconnectivity exposes networks to various security threats, including data interception, unauthorized access, and cyber-attacks. Hence, securing data transmission between these offices is of paramount importance.Network security is critical in protecting sensitive business information from cyber threats. A secure network ensures that data transmitted over the network is encrypted, authenticated, and protected against interception and tampering. This is especially crucial for businesses that deal with confidential customer information, financial transactions, and intellectual property.

Purpose of the Project

The purpose of the above project is to establish a secure communication channel between two internal networks (192.168.1.0/24 and 192.168.3.0/24) using an IPSec tunnel. This tunnel encrypts data packets as they travel between the networks over the internet (likely through the ISP), protecting sensitive information from unauthorized access.

Relevance to the Course

IPSec tunnel configuration plays a vital role in Digital Communication and Networks (DCN) for several reasons. Firstly, it underpins data center security. By understanding how to configure these tunnels, network professionals can safeguard sensitive information traversing potentially insecure networks within and between data centers. This ensures data confidentiality and integrity. Secondly, IPSec tunnels enable secure connectivity across geographically separated data center sites or between a data center and remote offices. This knowledge is crucial for building robust and secure data center networks, allowing for seamless communication and resource sharing across locations. Finally, familiarity with IPSec configuration empowers network engineers to troubleshoot connectivity issues within the data center network more effectively. By understanding the protocols and components involved, they can pinpoint and resolve problems more efficiently, maintaining optimal network performance.

Methodology

1. Basic Network Configuration:

- Hostnames: Assign unique hostnames (R1, ISP, and R3) for easy identification.
- Interfaces: Configure IP addresses and subnet masks for each router interface:
  - R1:
    - GigabitEthernet0/1 (internal): 192.168.1.1/24 (connects to the internal network)
    - GigabitEthernet0/0 (external): 209.165.100.1/24 (connects to the ISP)
  - ISP:
    - GigabitEthernet0/1 (internal): 209.165.200.2/24 (connects to R1)
    - GigabitEthernet0/0 (external): (IP address likely provided by your ISP, not configured here)
  - R3:
    - GigabitEthernet0/1 (internal): 192.168.3.1/24 (connects to the internal network)
    - GigabitEthernet0/0 (external): 209.165.200.1/24 (connects to R1)
- Routing: Configure a static route on R1 to send traffic destined for 192.168.3.0/24 (R3's internal network) to the ISP's interface (209.165.100.2). This ensures packets reach R3 through the tunnel.

2. Security License (Optional):

- The provided code snippet (license boot module c1900 technology-package securityk9) enables a security license (if applicable to your router model). Check your router's documentation for specific license requirements.

3. IPSec Tunnel Configuration (Crucial Step):

- ISAKMP Policy: Define an ISAKMP policy (e.g., 10) on both R1 and R3 for secure key exchange:
  - encryption aes 256: Use strong AES 256-bit encryption for data confidentiality.
  - authentication pre-share: Employ pre-shared keys for authentication (replace secretkey with a strong, shared secret).
  - group 5: Select a Diffie-Hellman group for key exchange (consult your router's documentation for supported groups).
- ISAKMP Key: Create a pre-shared key on both R1 and R3 for ISAKMP authentication:

    - crypto isakmp key secretkey address <peer_router_ip>: Replace <peer_router_ip> with the IP address of the other router (e.g., 209.165.200.1 on R1 and 209.165.100.1 on R3).
- IPsec Transform Set: Define an IPsec transform set (e.g., R1-R3) on both R1 and R3 to specify encryption and hashing algorithms:
    - esp-aes 256: Use AES 256 for data encryption within the tunnel.
    - esp-sha-hmac: Utilize SHA-hmac for message integrity and data authentication.
- IPsec Map: Create an IPsec map (e.g., IPSEC-MAP 10) on both R1 and R3 to link the ISAKMP policy, transform set, and traffic selectors:
    - set peer <peer_router_ip>: Specify the peer router's IP address.
    - set pfs group5: Enable Perfect Forward Secrecy (PFS) using the same Diffie-Hellman group (group 5) for additional security.
    - set security-association lifetime seconds 86400: Set the maximum lifetime of the security association (SA) to 24 hours.
    - set transform-set R1-R3: Assign the R1-R3 transform set for encryption and hashing.
    - match address 100: Create an access list (e.g., 100) to define the traffic that will be tunneled. The provided code permits all traffic between the internal networks (192.168.1.0/24 and 192.168.3.0/24).

Project Description

Nature of the Project

This project simulates securing a remote branch office within a distributed business environment. The focus lies on configuring a secure connection, specifically a Virtual Private Network (VPN), between the main office and the branch location. This VPN utilizes an IPSec tunnel to encrypt data transmissions, essentially creating a private network over the public internet. This project demonstrates a comprehensive approach to network security, ensuring data confidentiality, integrity, and availability while enabling seamless communication and collaboration across geographically dispersed teams.

Goals of the Project

- Secure communication between two internal networks despite being physically separated and connected through the internet.
- Implement robust encryption (AES 256) to ensure data confidentiality.
- Utilize pre-shared key authentication to verify the identity of communicating routers.
- Employ SHA-hmac for data integrity to detect and prevent message tampering.

Objectives of the Project

The objective of this project is to configure a secure connection between a main office and a remote branch office using Cisco Packet Tracer, a network simulation tool. This secure connection will protect data transfers from unauthorized access and ensure that the communication between the two offices is encrypted and authenticated

Specific Tasks Performed

This project creates a secure "highway" (IPsec tunnel) between two separate internal networks (like different office buildings). Data traveling on this highway gets scrambled (AES 256 encryption) to keep it confidential, even if someone snoops on the internet. To ensure only authorized "vehicles" (routers) can use the highway, they need a secret password (pre-shared key) to verify their identity. Additionally, a special security check (SHA-hmac) is performed to make sure the data hasn't been tampered with during its journey.

Materials Used

Software

Cisco Packet Tracer

Methods Employed

- **Planning and Design:** Defining the security requirements, network topology, and access controls for the VPN connection between the main office and the remote branch.
- **Configuration:** Setting up the VPN functionality on network devices (routers or firewalls) at both locations. This includes defining encryption algorithms, authentication methods, key exchange protocols (specific to IPSec), and routing protocols to ensure data reaches its intended destination.
- **Testing and Verification:** Validating the successful establishment of the secure tunnel, data encryption, and user access to shared resources across the VPN connection.

Code:

1. Starting configurations for R1, ISP, and R3. Paste to global config mode :

```
hostname R1
interface g0/1
ip address 192.168.1.1 255.255.255.0
no shut
interface g0/0
ip address 209.165.100.1 255.255.255.0
no shut
exit
ip route 0.0.0.0 0.0.0.0 209.165.100.2

hostname ISP
interface g0/1
ip address 209.165.200.2 255.255.255.0
no shut
interface g0/0
ip address 209.165.100.2 255.255.255.0
no shut
exit

hostname R3
interface g0/1
```

```
ip address 192.168.3.1 255.255.255.0
no shut
interface g0/0
ip address 209.165.200.1 255.255.255.0
no shut
exit
```

2. Make sure routers have the security license enabled:

```
license boot module c1900 technology-package securityk9
```

1. Configure IPsec on the routers at each end of the tunnel (R1 and R3):

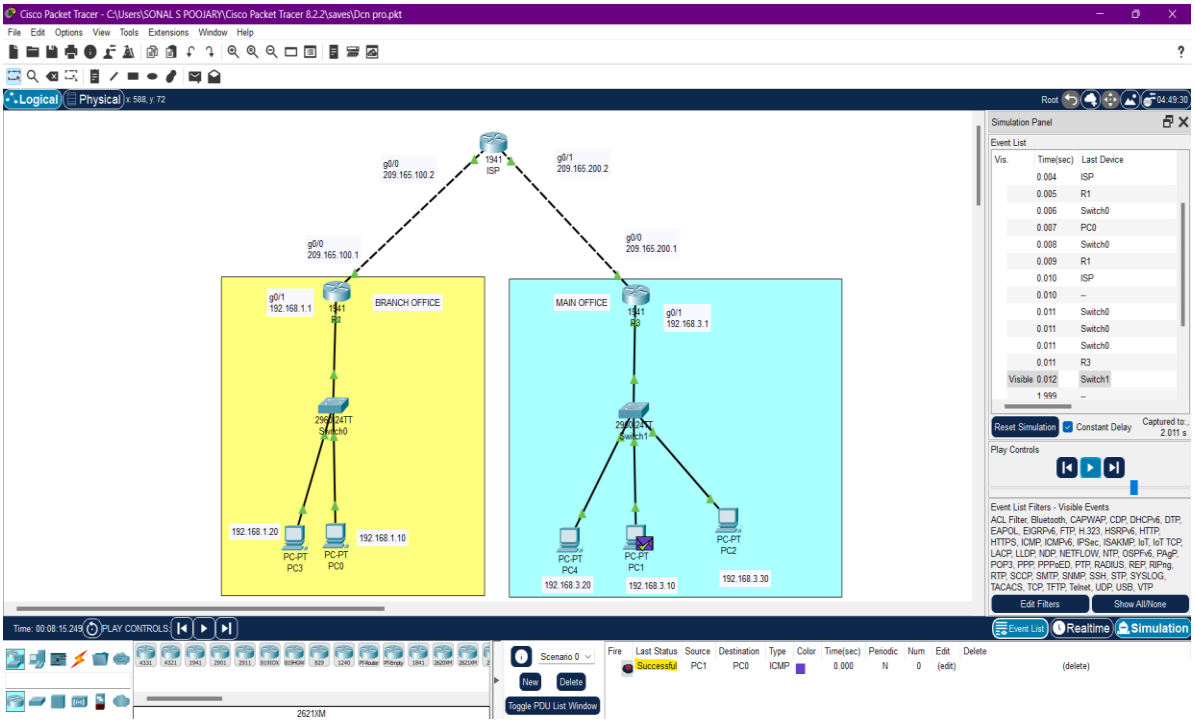
```
!R1
crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 5
!
crypto isakmp key secretkey address 209.165.200.1
!
crypto ipsec transform-set R1-R3 esp-aes 256 esp-sha-hmac
!
crypto map IPSEC-MAP 10 ipsec-isakmp
set peer 209.165.200.1
set pfs group5
set security-association lifetime seconds 86400
set transform-set R1-R3
match address 100
!
interface GigabitEthernet0/0
crypto map IPSEC-MAP
!
access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
```

```
!R3
crypto isakmp policy 10
encryption aes 256
authentication pre-share
group 5
!
crypto isakmp key secretkey address 209.165.100.1
!
crypto ipsec transform-set R3-R1 esp-aes 256 esp-sha-hmac
!
crypto map IPSEC-MAP 10 ipsec-isakmp
set peer 209.165.100.1
```

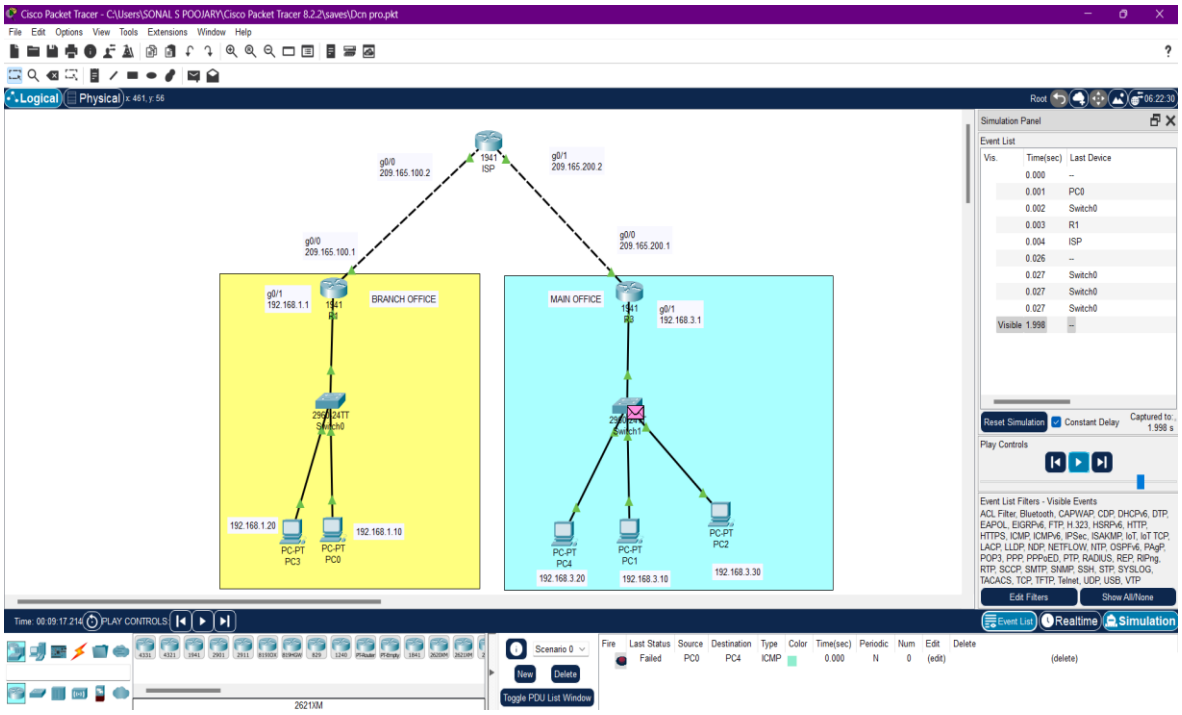
```
    set pfs group5
    set security-association lifetime seconds 86400
    set transform-set R3-R1
    match address 100
    !
    interface GigabitEthernet0/0
    crypto map IPSEC-MAP
    !
access-list 100 permit ip 192.168.3.0 0.0.0.255 192.168.1.0 0.0.0.255
```

Results and Analysis

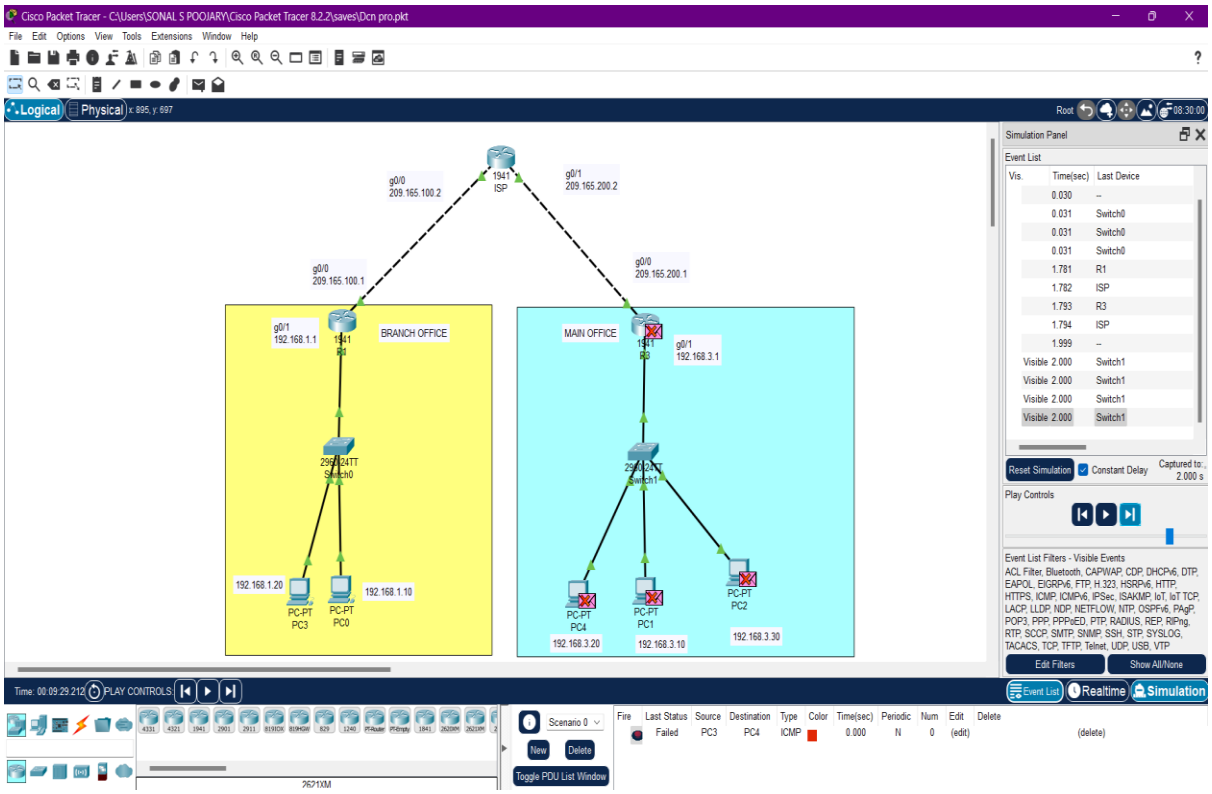
Secure-to-Secure Transfer(from PC0 to PC1):



Secure-to-Unsecure Transfer(from PC0 to PC4):



Unsecure-to-Unsecure Transfer(from PC3 to PC2):



Analysis:

- It encrypts data (AES 256) for confidentiality and verifies router identity (pre-shared key) for secure communication.
- Encryption adds some strain on network performance, but the benefits outweigh the drawbacks.
- Optimizations can be achieved by focusing encryption on critical traffic and implementing a robust key management system.
- Exploring Perfect Forward Secrecy can further enhance security.

Conclusion

The project of securing a remote branch office by configuring a secure connection to the main office using Cisco Packet Tracer demonstrates a comprehensive approach to enhancing network security in a distributed business environment. The core element is establishing a Virtual Private Network (VPN) to safeguard data traveling between the main office and the branch. This VPN, typically an IPSec tunnel, encrypts data transmissions, creating a secure pathway over the public internet. By acting as a private network within the internet, the VPN shields data from unauthorized access, ensuring confidentiality, integrity, and availability. This secure connection allows authorized users in both locations to seamlessly access shared resources and collaborate as if they were physically on the same internal network. This approach not only bolsters overall network security but also fosters improved communication and productivity across geographically separated teams.

References

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